Next-Gen AI Compiler

Agents On The Control Plane

Agents Own Search · Orchestration · Synthesis.
Compilers Own Lowering · Legality · Measure · Fallback.

Ask: Is The Hybrid Prediction (∼2027–31) The Right Bet For Your Roadmap?

Expert Briefing · ∼25–30 min + Q&A · Prediction-first

Agenda

1. Trends · Keep Substrate
2. Verdict + Four Jobs
3. Architecture Evolution
4. Prediction Checkpoints
5. Commercial Blockers + Gaps
6. Stance · Checklist · Ask

Trend backdrop — two stacks converging

Compilers for AI
mature · fragmented substrate

AI for compilers
rapid hybrid growth

Next-gen ≠ throw away LLVM/MLIR — make them **agent-addressable**.

Six active trends

A — Hybrid guidance
Free IR rewrite is fragile
(mlirAgent < identity)
Constrain the action space:
advisory metadata · validated hints
pass lists for `opt` to apply
AgentCompile / HintPilot / LLM Compiler

B — RL → agents
From CompilerGym policies to
tool-using / multi-agent loops
Heuristic synthesis as shippable C++
Workflow compile · freeze · place
Compiler-R1 / Magellan / FlowCompile

C — MLIR + Triton substrate
PyTorch → Inductor → Triton
Parallel: StableHLO → XLA / IREE
Peak often still vendor libs
Tile / CUTLASS / FlashAttention
MLIR shared; product paths diverge

D — Kernel agents industrial
KernelBench: correct *and* faster
One-shot often <20%; fusion hard
GEAK / KernelLLM / AgentCompile
Refinement ↑ correctness, not always speed
Bottleneck for new models + portability

E — Verify in the loop
Unit / golden → numerical checks
→ Alive2-class local formal
Strong locally; weak on GPU races
and floating-point nondeterminism
Admit needs a stacked oracle ladder

F — Broader compile object
Mid-decode diagnosis (generative)
FMware: prompts / agents / knobs
Agents as compiler engineers
Heuristics rewritten in-tree
Compiler.next / Magellan / Claude C

Keep substrate, change control plane

Preserve

deterministic lowering
library kernels
build-CI regressability
local formal checks

Adopt

advisory search
multi-agent orchestration
heuristic synthesis
profile / verifier feedback

Anti-pattern: replace `opt`/Inductor with unconstrained LLM codegen

Executive verdict

Agents reshape the control plane more than they replace the data plane.

Control plane = search / decide layer

Agents own: search, orchestration, synthesis

propose → measure → admit

Data plane = lower / execute layer

Compilers own: lowering, legality, measure, fallback

Inductor / MLIR / Triton stay the default path

Same stack, different jobs — agents above classical lowering, not instead of it.

Compilers for AI × AI for compilers

↓

Hybrid LLM–compiler loops

↓

4th job Tier A: bring-up / codesign

Will not happen soon

- Unconstrained LLM replaces Inductor / `opt`
- Autonomous chip tape-out via compiler agents (C10)

Four agent jobs — architecture spine

(a) Online
propose → measure admit
CompileIQ / GEAK
ACCLAIM / HintPilot

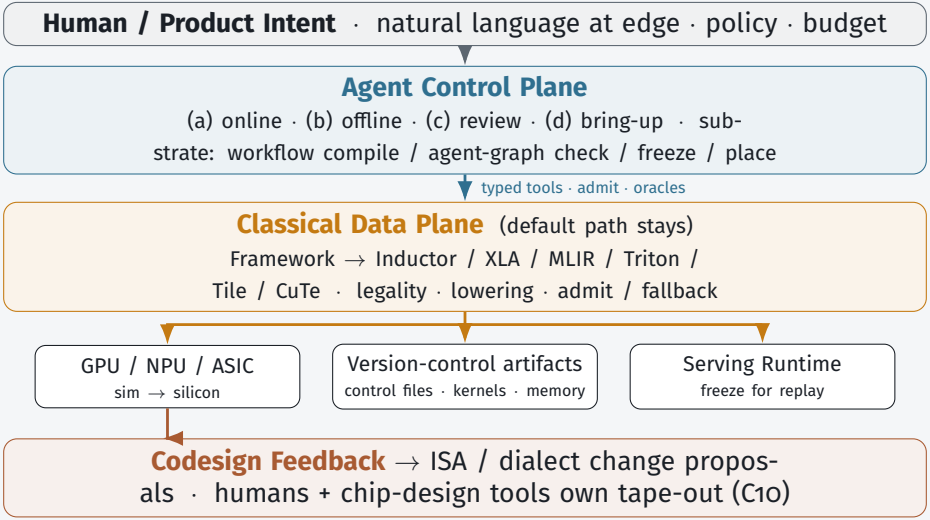
(b) Offline
evolve heuristics → C++ / MLGO
Magellan
AlphaEvolve

(c) Eng.
oracle-gated pull request / change
Archer

(d) Bring-up
coverage → performance
sim + silicon
TritorX
KernelEvolve

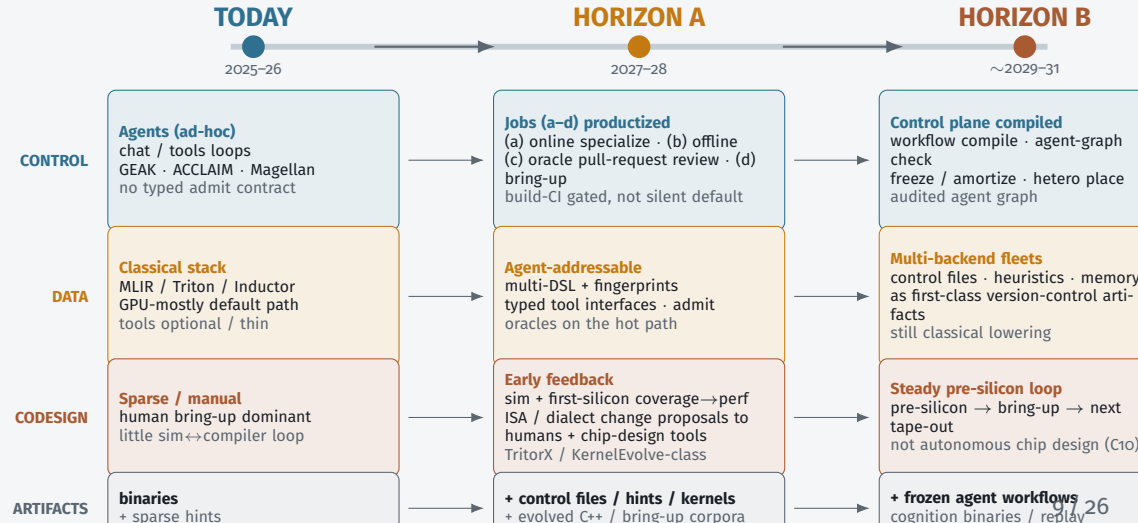
Control plane jobs sit *above* classical lowering — not instead of it.

Architecture — Target Stack



Invariant: LLM guides search — does not silently define unchecked executable behavior.

Architecture evolution — component changes



What ships / does not by 2028

Predicted to ship

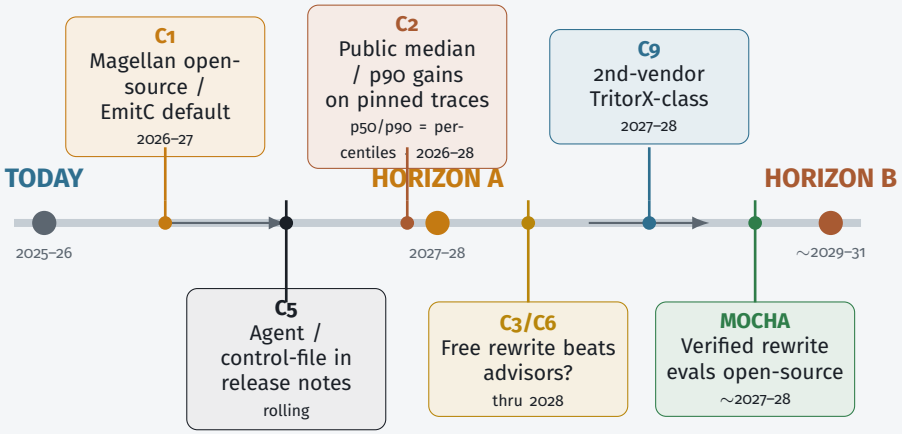
- Agent-addressable tool interfaces
- Hot-path specialize (not silent default-all)
- Magellan *and* MLGO both live
- Oracle pull-request review in serious orgs
- Coverage-first ASIC bring-up
- Triton-family primary; multi-DSL rising

Does *not* ship

- Unconstrained LLM replaces `opt`/Inductor
- One agent IR for all vendors
- Kernel agents uniformly beat eager on fusion ladders
- Autonomous microarch tape-out

Horizon A success = build-CI gated specialize + oracles + freeze artifacts

Roadmap Checkpoints — when the prediction changes



Falsifiable milestones — watch settlement signals; do not average disagreements.

Checkpoint C1 — heuristics vs neural advisors

A — Magellan path

evolve shippable C++ heuristics

Evolve shippable C++ (EVOLVE-BLOCK)

Offline coding agent *is* the control-plane output

Product = evolutionary coding agent + review

B — MLGO / EmitC path

MLGO = learned in-tree LLVM advisors

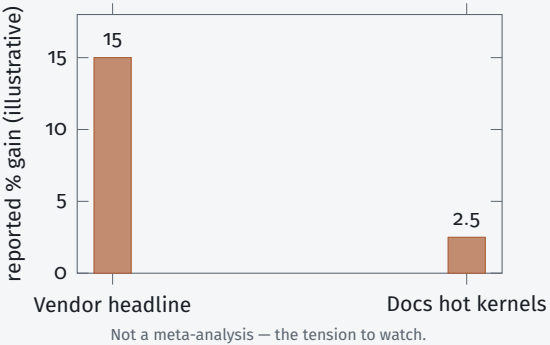
NN advisors stay in-tree

June 2026 plan of record: internal inliner → Android/Fuchsia → Chrome multi-model

Agents train/feature NNs

Settlement: public Magellan llvm patches that displace MLGO *or* EmitC-MLGO becomes customer default
Watch LLVM syncs + OpenEvolve recipes quarterly through 2027

Checkpoints C2 + C5 — gains & default path



Not a meta-analysis — the tension to watch.

C2 "Agents become default" needs **median (p50) build-CI wins**, not best-kernel blogs.

p50 / p90 = 50th / 90th percentile of the gain distribution across builds.

C5 Online specialize (control files / hints) vs offline eng (Magellan/MLGO) — both may live; *which is the default flag?*

Settlement: pinned public traces + release notes listing agent / control-file workflows.

Checkpoints C3 / C6 / C9 / C10 — interface width & scope

C3 — free rewrite vs advisory
Flip: shared suite where free rewrite wins
⇒ wide rewrite interface vs narrow control files / hints

C6 — replace vs control plane
Flip: production default with *no* admit
⇒ agents replaced the compiler (we reject this)

C9 — coverage vs peak
Flip: TritorX → KernelEvolve ladder public
⇒ SKU: coverage service target then performance target

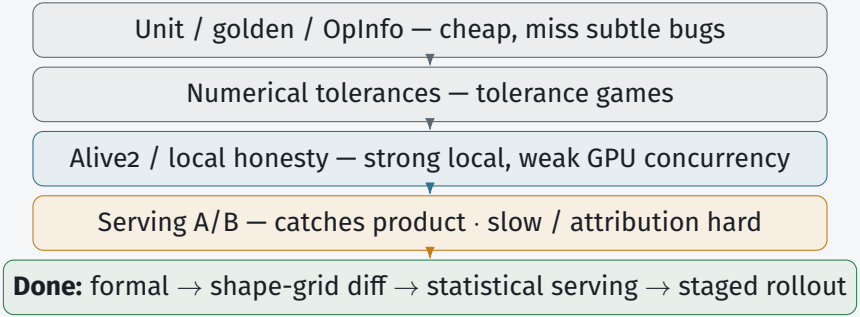
C10 — codesign vs auto tape-out
Flip: microarch primarily agent-proposed *and* compiler-oracle validated ⇒ enters chip-design tooling

Commercial blockers — top 5

1. **Oracles strong enough for money**
   Fast-but-wrong · liability
2. **Cost / replay / when agents run**
   Nondeterministic $N compiles
3. **Agent↔compiler contract**
   Natural-language-only = demo
4. **Distributional production evidence**
   No default-path A/B
5. **Ownership · supply chain · human review**
   Unowned agent code in trusted base

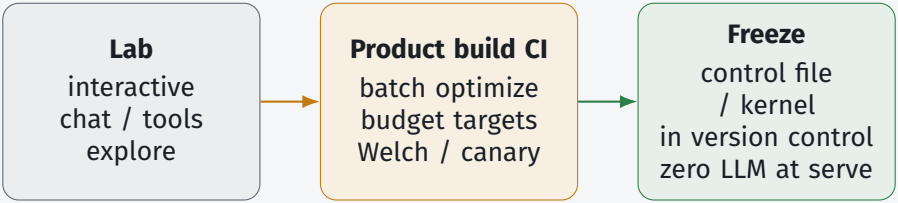
Each blocker gets one slide — productization gaps, not research wishlist.

Blocker 1 — Oracles for money



Room question: who owns false negatives when admit passes and production miscompiles?

Blocker 2 — Cost, replay, when may the agent run?



Cache key: (IR hash, hardware, compiler ver, agent policy) · budget $/%gain

Falsifies commercially: “chat with compiler on every build” without spend/latency targets.

Blocker 3 — Agent↔compiler contract

A. Natural language + pasted logs — demo only

B. Structured admit traces — build CI / bill of materials

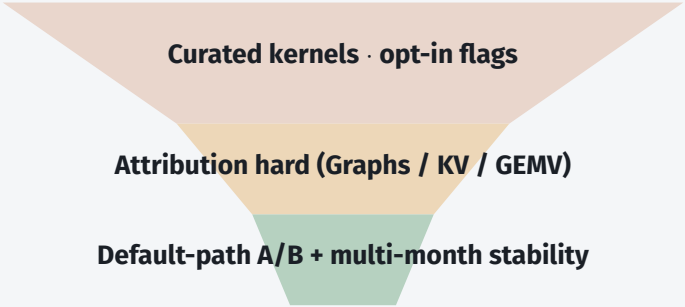
C. Typed tool interfaces — required action space

D. Hybrid — natural-language view over traces

```
admit_record = {graph_hash, hw_id, compiler_ver, action[], oracle[],
artifact_digest, policy_id}
```

Lean: **C + B**; Natural language is a view, not source of truth. Advanced Control File = portable compiler knobs.

Blocker 4 — Distributional production evidence



Marketing bar: median + 90th-percentile gains + cost-to-compile · persistent serving throughput, not single-kernel headlines

Blocker 5 — Ownership, security, human review

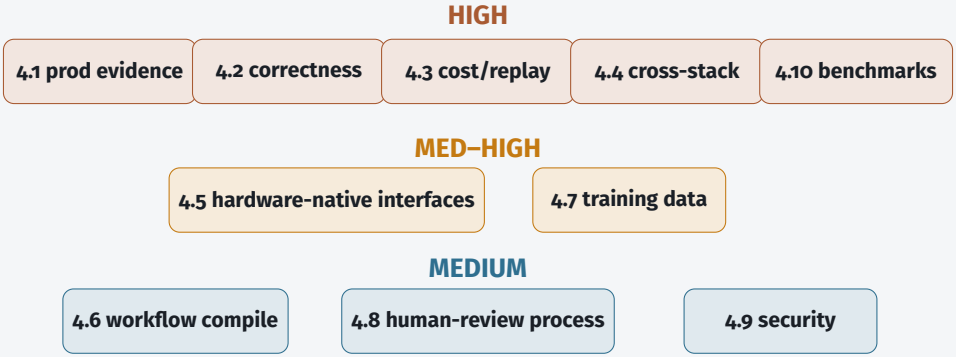
Trusted-base risk
agent kernels / heuristics enter trusted base

Ownership
CODEOWNERS
signed provenance
sandbox

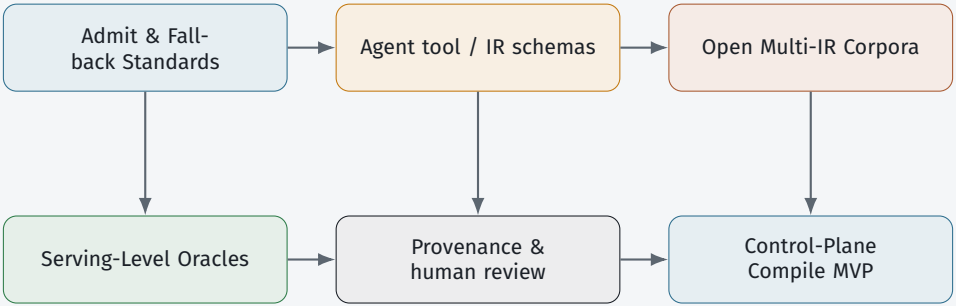
Human-review capacity
agents ↑ drafts
reviewers = bottleneck

Lean: human CODEOWNER + signed admit + sandbox · oracle auto-merge only for **narrow** action classes
Generic forge AI ≠ compiler-oracle review (C7)

Gap map — what blocks the prediction, not a wishlist



Cross-Cutting Research Agenda



Near-Term: Specs That Make Hybrid Pipelines Build-CI Regressable.

Org Adoption Questions

1. Online (CompileIQ) vs Offline (Magellan) — which matches your release model?

2. Oracle stack: Alive2 / golden / serving A/B — who owns false negatives?

3. Agent contract surface: LLVM / MLIR / Triton / StableHLO / Tile?

4. How do you cache/regress traces across compiler *and* model upgrades?

5. Max $/build for a median X% win? Named maintainer per admitted artifact?

Working stance until conflicts settle

1. Hybrid: agents search; compilers lower (not replace)
2. Magellan || MLGO — parallel bets (C1)
3. Discount single-number speedups (C2)
4. Constrained actions + strong oracles (C3)
5. Demote generic SCM AI (C7)
6. Coverage→performance ladder; no autonomous chip design (C9/C10)

Commercial checklist — handout

Architecture

typed tools + admit traces
version control ≫ dense memory ≫ scratchpad
state-machine plan for service targets
freeze before default-on

Trust

layered oracles
CODEOWNERS + provenance
joint version pins
A/B before “prod default”

Business

sell regressable artifacts + build-CI quota
customer owns outputs
coverage then performance service target
token budget per %gain

Discussion

Thank you — hybrid bet stays the ask

1. Which checkpoint flips your investment first — C1, C2, or C9?
2. Harder commercial bar: **oracle strength** or **replay/freeze**?
3. Building job (a), (b), or (d) first — what do you refuse online?

Ask again: is the hybrid bet right for your roadmap? Watch settlement signals — do not average disagreements.